



Molecular dynamics simulation of hyaluronic acid hydrogels: Effect of water content on mechanical and tribological properties

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ABSTRACT

Background and objective: Recently conducted biomedical studies have shown that the drug diffusivity of hyaluronic acid hydrogel plays an important role in the treatment of joint diseases. The drug diffusivity is closely related to the water content of hydrogel. In addition, different water content will not only affect its mechanical and tribological properties, but also change the effect of drug release.

Methods: In this work, a Molecular dynamics simulation was used to investigate the effect of water content on spatial distribution, tribological and mechanical properties of a hyaluronic acid hydrogel network. This paper focuses on the analysis and calculation of the radial distribution function of 20, 40, 60, and 80% water content model and the friction force and mechanical parameters under the influence of different load and friction speed.

Results: The results show that at 20 and 40% water content, the spatial distribution is loose and the intermolecular force is not strong, resulting in a major lack in tribological and mechanical properties; whereas at 60 and 80% water content, the spatial distribution becomes gradually compact and the intermolecular force is gradually increased. The tribological and mechanical properties manifest a marked improvement. **Conclusions:** The calculations reveal that the hydrogel model has the best wear resistance, pressure resistance, and plastic deformation resistance at 80% water content. In the range of 20–80% water content, the mechanical properties and friction properties of hydrogels become better and better with the increase of water content.

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1. Introduction

Hydrogels are water-containing materials with good biocompatibility, durability, and other unique properties after cross-linking with other substances, and have become more popular in tissue engineering [1–3], drug sustained-release [4–6], disease treatment [7,8], and many other fields. Natural hydrogels have attracted wide attention in simulating tissue structure [9], cell proliferation and differentiation [10–12] and substance transport [13,14]. Researchers are increasingly resorting to Hyaluronic acid (HA) hydrogel as a natural hydrogel with enormous potential. HA is a component of extracellular matrix and synovial fluid which are both abundant in articular cartilage and its biocompatibility and biodegradability render it very useful in tissue engineering applications [15–17], bone regeneration [18–20], and joint disease treatment [21,22].

Hydrophilic HA hydrogel can absorb substantial amounts of water [23–25]. The amount of water in the hydrogel has a direct influence on its performance. As a method for treating retinal ailments, Dromel et al. [26] examined how varying water content affects medication release from injectable HA hydrogels. The degree of drug molecular diffusion of drug-loaded hydrogels increases with water content, as per a study comparing drug molecular diffusion of drug-loaded hydrogels. Veskovic et al. [27] employed EPR to track drug release, swelling, and breakdown in various hydrogels. When Tran et al. [28] studied the mechanical properties of composite hydrogels with high water content. Young's modulus and tensile strength of hydrogels with different water content were tested to speculate the best water-bearing state.

Many researchers have studied extensively on the macroscopic mechanical properties of hydrogels. To study and analyze the mechanical and friction behavior of hydrogels with different water content, it is also very important to study the effect of water molecular weight on the properties of hydrogels at the molecular level. All-atomic molecular dynamics (MD) simulation technology

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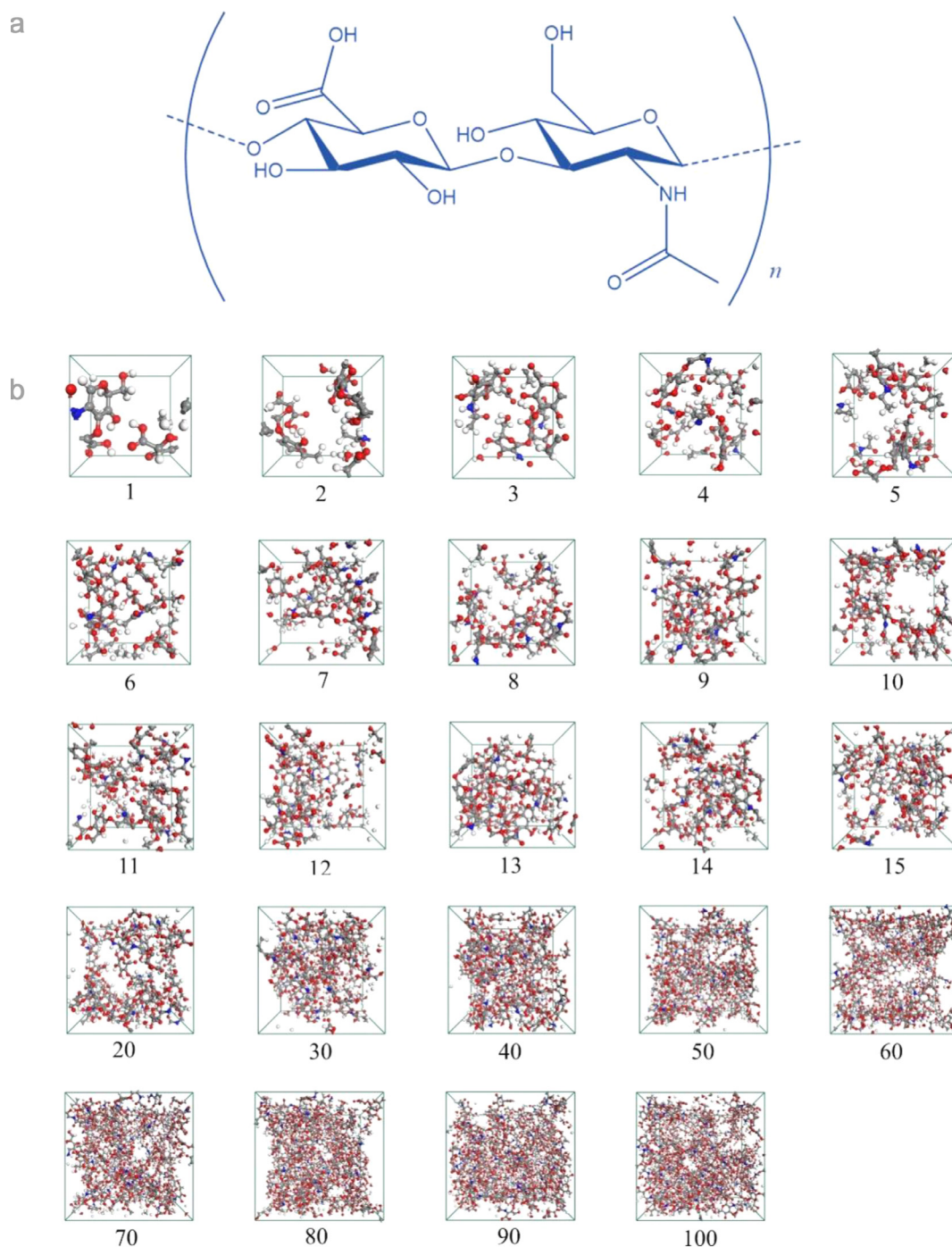


Fig. 1. (a) HA unit molecular structure; (b) DP is 1–100 HA model; (c) DP is 1–100 model CED curve; (d) finalize the HA model of DP = 50 used. Black represents carbon atom, red represents oxygen atom, dark blue represents nitrogen atom, and white represents hydrogen atom.

has evolved into a potent scientific research tool for building, displaying, and analyzing structural models of molecules, solids, and surfaces, as well as studying their properties. It provides a clear grasp of polymer molecular structure. Micromechanics and macroscopic experimental data from molecular dynamics simulations are combined to provide a more comprehensive explanation of material properties [29–32].

Sun et al. [29] used molecular dynamics simulations to study the effects of different water content on the network structure and oxygen permeability of poly(bis(trimethylsilyloxy)methylsilylpropyl glycerol methacrylate) (PSiMA) and poly(2-methacryloyloxyethyl

phosphorylcholine) (PMPC) hydrogels. The results show that an increase in hydrogel water content is closely related to a decrement in its oxygen permeability. Lee et al. [33] employed an all-atomic MD simulation to investigate the mechanical characteristics and equilibrium structure of block and random sequence models of Poly (N-vinyl-2-pyrrolidone-co-2-hydroxyethyl methacrylate) hydrogels under varied water content. MD simulations of gel self-healing behavior [31], drug molecular transport [34], molecular permeability [35], shock response [36], and hydrogel deformation [37] were also performed. Friction is both an ancient and a young science [44]. In current MD simulation-based research, there are

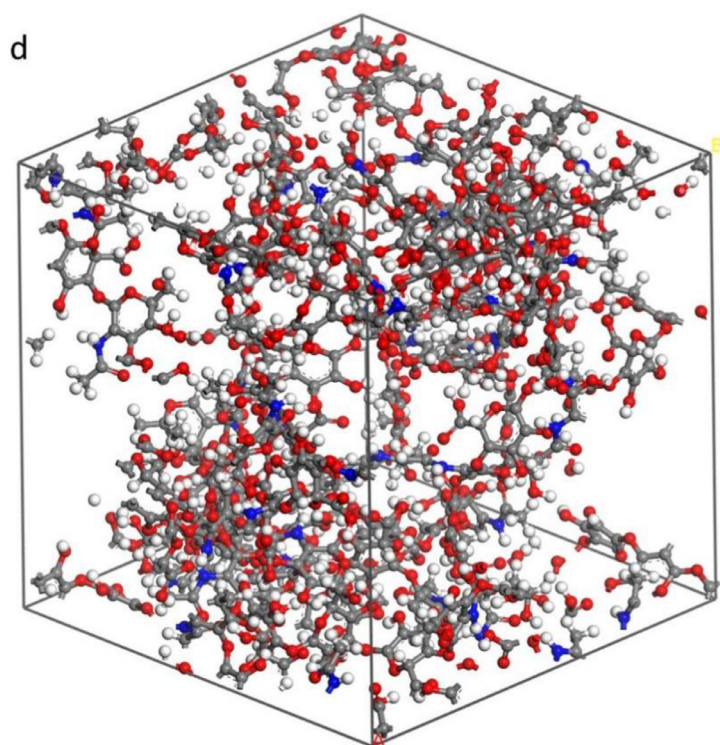
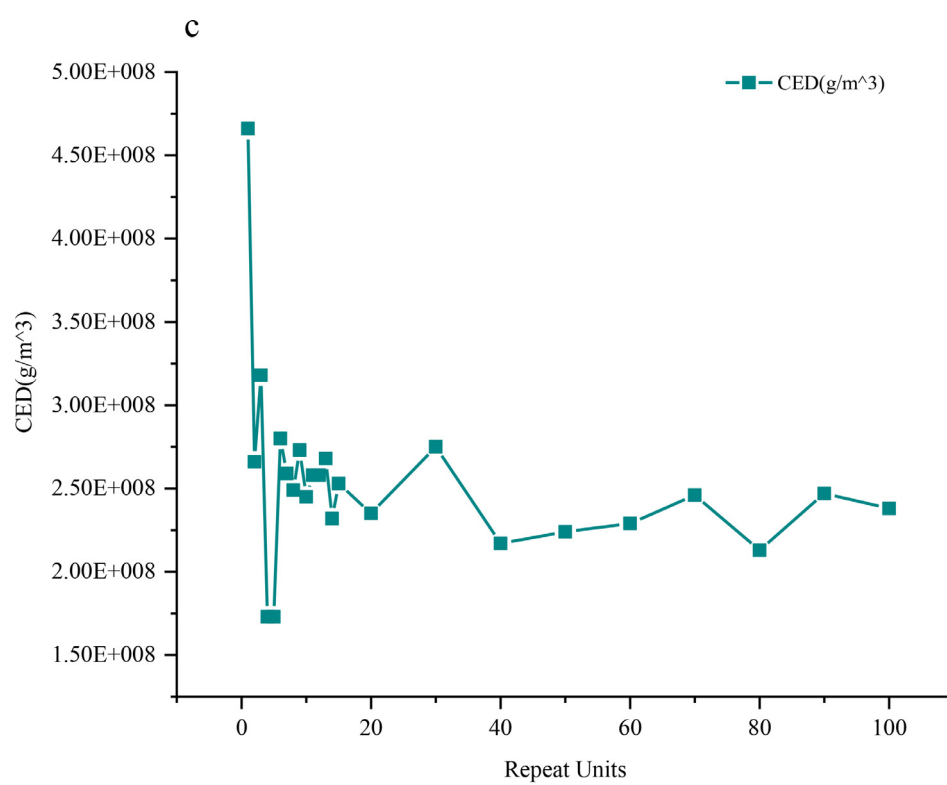
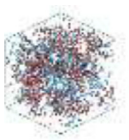
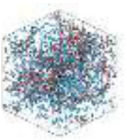
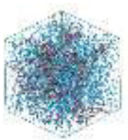


Fig. 1. Continued

Table 1
Characteristics of hydrogel models with different water content (light blue represents water molecules).

Water content	20wt%	40wt%	60wt%	80wt%
				
3D model				
Number of HA chains	1	1	1	1
HA Number of repeated units	50	50	50	50
Bias temperature (K)	298	298	298	298
Number of water molecules	258	687	1546	4122
3D model size (Å)	33.779773	37.175826	42.557054	53.616769
Boundary conditions	periodic	periodic	periodic	periodic

only a few investigations on the mechanics and wear resistance of hydrogels with varied water content at the molecular structural level, and this subject remains largely untouched.

The influence of water content on the mechanical characteristics and wear resistance of hyaluronic acid hydrogels was investigated and analyzed in this paper. All-atomic molecular dynamics (MD) simulation was employed to simulate the network of hyaluronic acid hydrogels with varied water content. The goal of this work is to investigate and analyze the molecular space distribution after different compression strains, the mechanical properties affected by different water content, and the wear resistance of hydrogels affected by different friction environments under the influence of different water content. Molecules and atoms' micro-scope states and influencing mechanisms are discussed in depth.

2. Methods

2.1. Model construction

2.1.1. Establish the foundation of the HA network

In this research work, the single molecular model of the hydrogel material is first established, and then the appropriate molecular chain is established from the monomer. The Amorphous models of water molecules and chains of hydrogel materials are established through the "Amorphous Cell" module.

The HA hydrogel network is composed of a large number of monomer sequence cycles [38]. In the model establishment stage, the cohesion energy density of the model with the degree of polymerization (DP) ranging from 1 to 100 is detected, as shown in Fig. 1b.

The size of the intermolecular force is usually expressed by the cohesive energy density. The cohesive energy density is defined as the energy required to overcome the intermolecular forces and move 1 mol of liquid or solid molecules to their intermolecular gravitational range per unit volume. The calculation formula is as follows, where CED is the cohesive energy density, E_{coh} is the cohesive energy, and V_{mol} is the molar volume.

$$CED = E_{coh}/V_{mol}$$

The cohesive energy density is the key to establishing the molecular chain with varying DP, however, the cohesive energy density of the HA molecular chain model of different DP will not be the same. Through the simulation calculation of several molecular chains with different degrees of polymerization, it can be seen that the fluctuation of the cohesive energy density of the model tends to be stable when DP begins at 17, as shown in Fig. 1c. Therefore, as long as DP is greater than 17, it can be employed as the best chain length of the model, and as shown in Fig. 1d, it is finally decided to assume a DP equivalent to 50 as the basis of the model.

Table 2
Parameter tables of Young modulus and Poisson ratios with different water content.

Physical quantity	Young Modulus(Gpa)	Poisson Ratios		
Water content 20wt%	X = 3.0837 Y = 3.4344 Z = 3.4207	Exy = 0.1292 Eyx = 0.1439 Ezx = 0.1278	Exz = 0.1152 Eyz = 0.1604 Ezy = 0.1597	
Water content 40wt%	X = 4.3023 Y = 3.9822 Z = 3.7627	Exy = 0.1493 Eyx = 0.1382 Ezx = 0.1217	Exz = 0.1392 Eyz = 0.2127 Ezy = 0.2009	
Water content 60wt%	X = 4.6263 Y = 4.6687 Z = 4.6524	Exy = 0.1960 Eyx = 0.1978 Ezx = 0.2600	Exz = 0.2586 Eyz = 0.2828 Ezy = 0.2819	
Water content 80wt%	X = 5.8769 Y = 5.6882 Z = 5.8033	Exy = 0.2885 Eyx = 0.2792 Ezx = 0.2897	Exz = 0.2933 Eyz = 0.2926 Ezy = 0.2985	

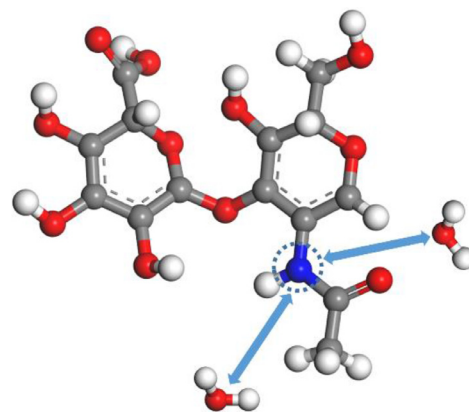


Fig. 2. $g(r)$ shows the radial distribution function of N atom and O atom of water molecule in HA molecular chain.

2.1.2. Establishment of hydrogel polymerization network

After selecting the HA model with a DP of 50, water molecules accounting for 20wt%, 40wt%, 60wt% and 80wt% of the whole network mass were added to form different water content models. All models are set to periodic boundary conditions. In this paper, it is assumed that all the atoms are within the boundary of the model, which makes it easier for the model to form a comparative reference while keeping the same shape. The data pertaining to the model is shown in Table 1.

The structure of the model network is random when the model with varying water content is constructed, the energy is not in a stable state, and the high energy of the model is not favorable to simulation analysis. Hence the model is geometrically optimized and annealed. In the process of geometric optimization calculation, the Compass force field is used for simulation calculation, the calculation quality is set to the best, and the Max iterations is set to

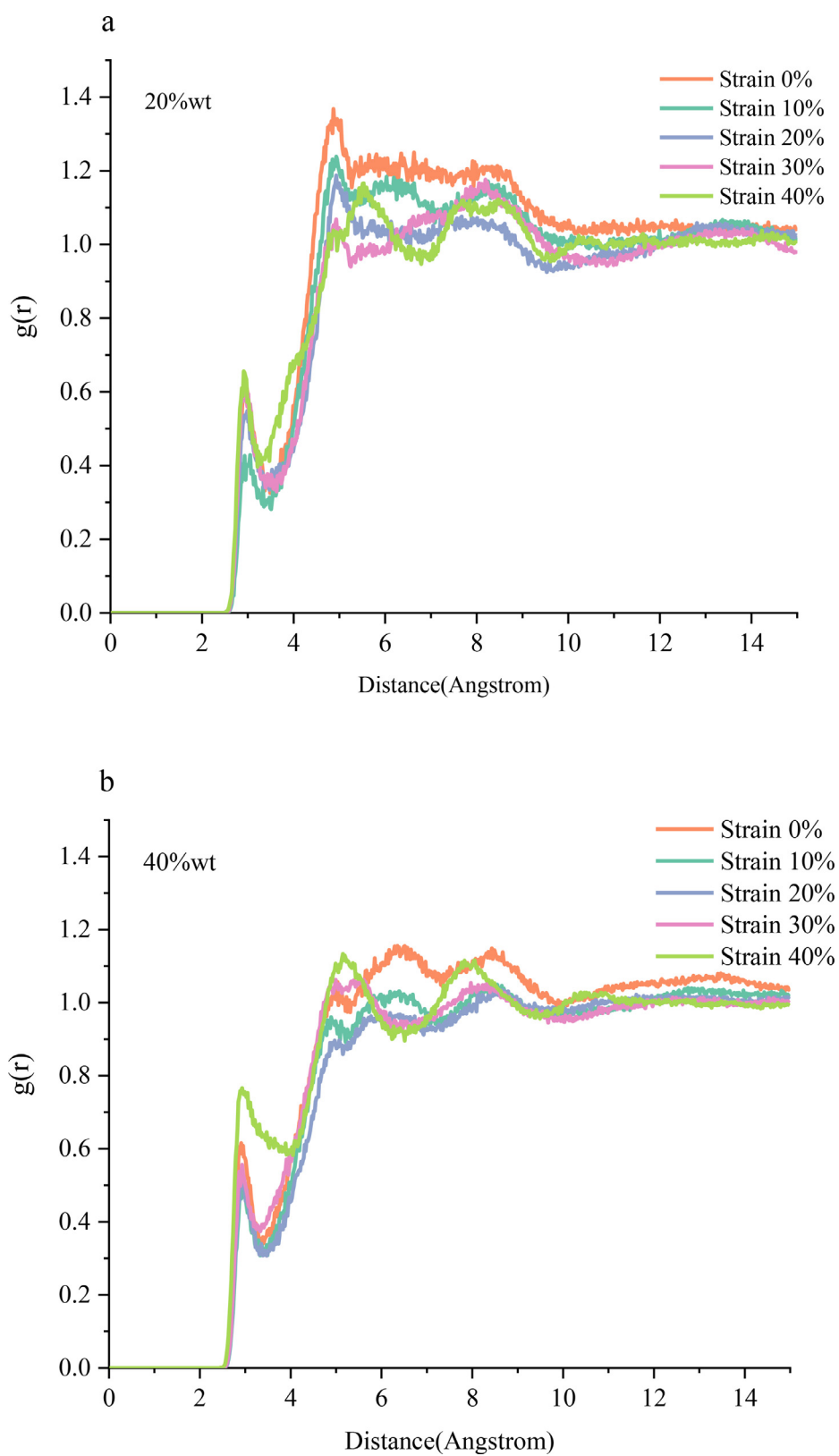


Fig. 3. The N(HA)-O(H₂O) radial distribution function of the hydrogel model with different water content when the compression variables are 20% (a), 40% (b), 60% (c) and 80% (d).

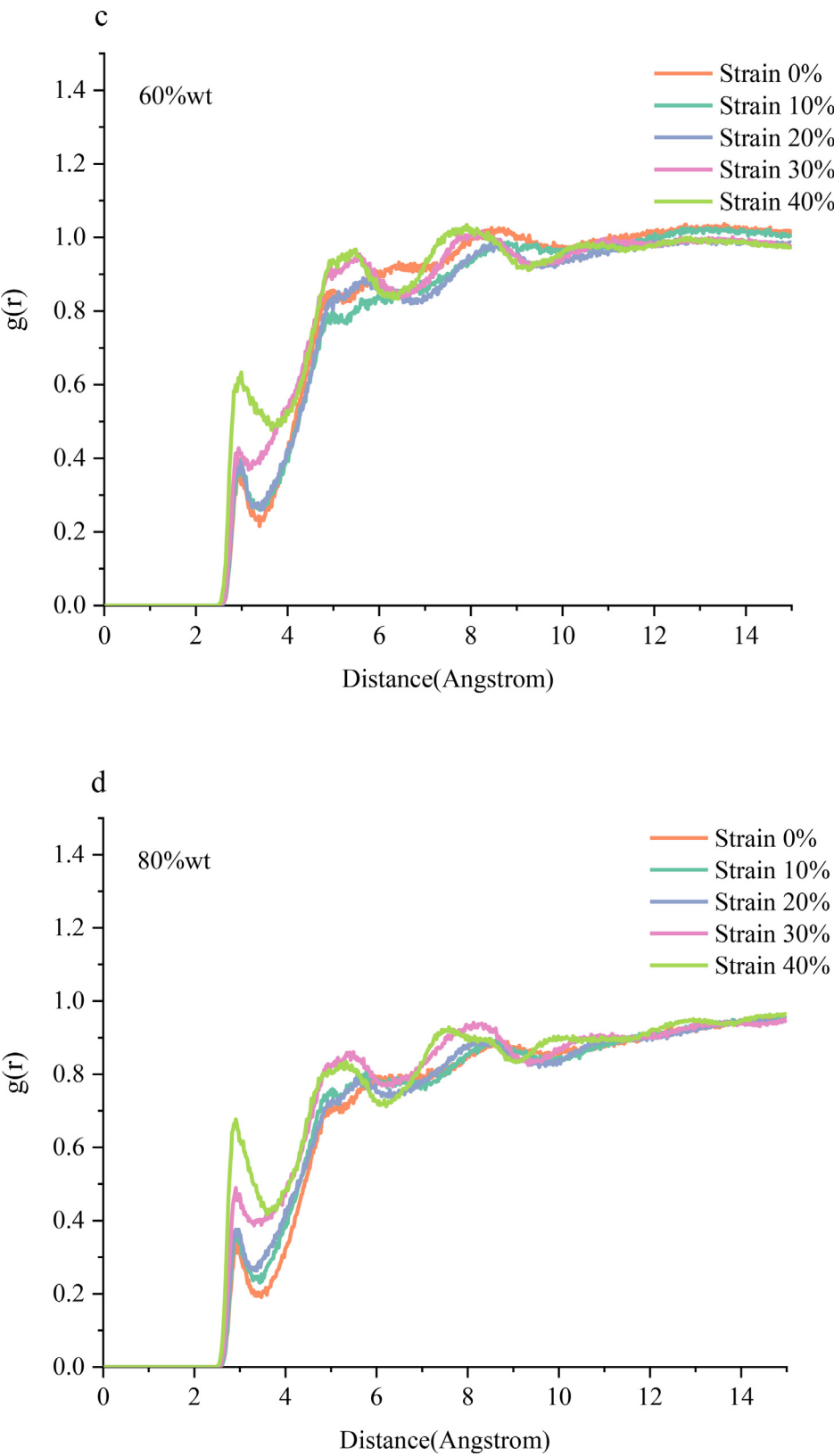


Fig. 3. Continued

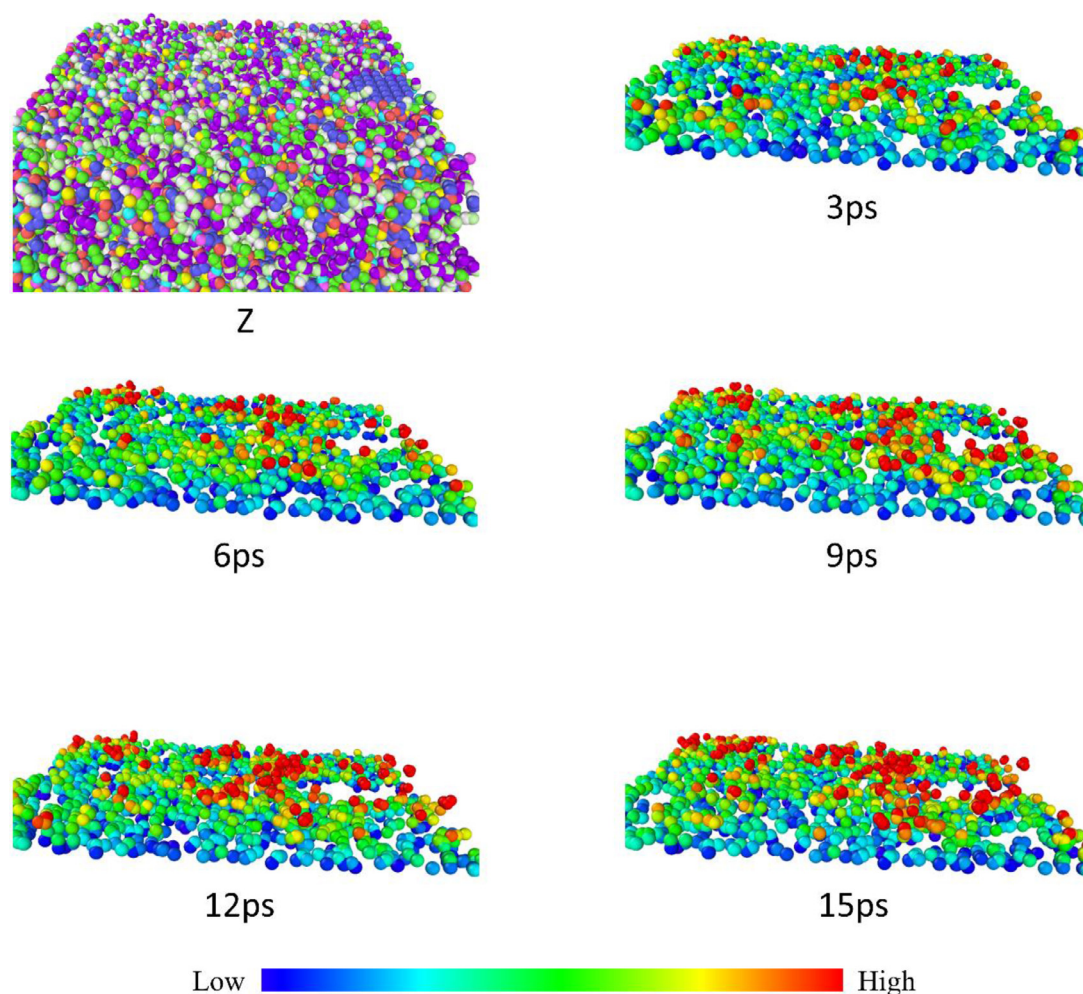


Fig. 4. The arrangement diagram of friction atoms with different water content. The 3 ps to 15 ps graphs show atomic stacking. The blue, green, yellow and red spheres in turn show that the relative height of the atom is getting larger and larger, and the highest point is red.

500,000, so that it can be fully calculated. When performing annealing calculations, the temperature of the model is reduced from 800 K to 300 K and then elevated from 300 K to 800 K. The process is repeated to keep the energy in a stable state. The purpose of this mode of operation is to get the model to equilibrium structure in a short time [33] and based on this stability, the MD simulation analysis is carried out.

2.2. MD simulates the settings of the main parameters

2.2.1. Selection of force field

To mimic the motion of the molecular system, MD mostly uses Newtonian mechanics. It takes samples from a system made up of multiple states of the molecular system, calculates the system's configuration integral, and then uses the configuration integral to compute the thermodynamic quantities and other macroscopic features of the system.

The force field is chosen early in the MD simulation process, and it is usually one of the five force fields available: CHARMM, CVFF, PCFF, and COMPASS. Small organic molecules, liquids, polymers, and biological systems can all benefit from the CHARMM force field. The CVFF force field was first used in the field of biochemistry. It has gradually begun to be applied to the simulation calculation of organic molecules after continual research and application and has furnished good results in the calculation of configuration and binding energy. Polymers and organic systems, as well

as inert gasses and a range of metal atomic force field parameters, can all be calculated using the PCFF force field. The COMPASS force field is the first to unite the organic molecular system's force field with the inorganic molecular system's force field. Before the emergence of the COMPASS force field, these two force fields were used individually. The categorization treatment is employed in the treatment of organic and inorganic systems, and different systems utilize different models; even for the two mixed systems, an acceptable model may be used to explain. Small molecules, polymers, metal ions, metal complexes, and metals can all be simulated using this force field. Because the friction calculation is carried out in the LAMMPS software and the mechanical calculation is carried out in the Materials Studio (MS) software, but the model derived from MS cannot use the compass force field in LAMMPS. After comparison, for mechanical and spatial distribution simulation, we use the compass force field, and for friction simulation, we use the PCFF force field.

2.2.2. Selection of ensemble

MD simulation employs statistical mechanics to compute the attributes of a system via a specific thermodynamic ensemble, which employs several algorithms to calculate energy, force, and molecular arrangement, among other things. The ensemble is a collection of systems with similar macro properties but differing micro properties that can measure all of the system's microstates. In this study, the model will be annealed and geometrically op-

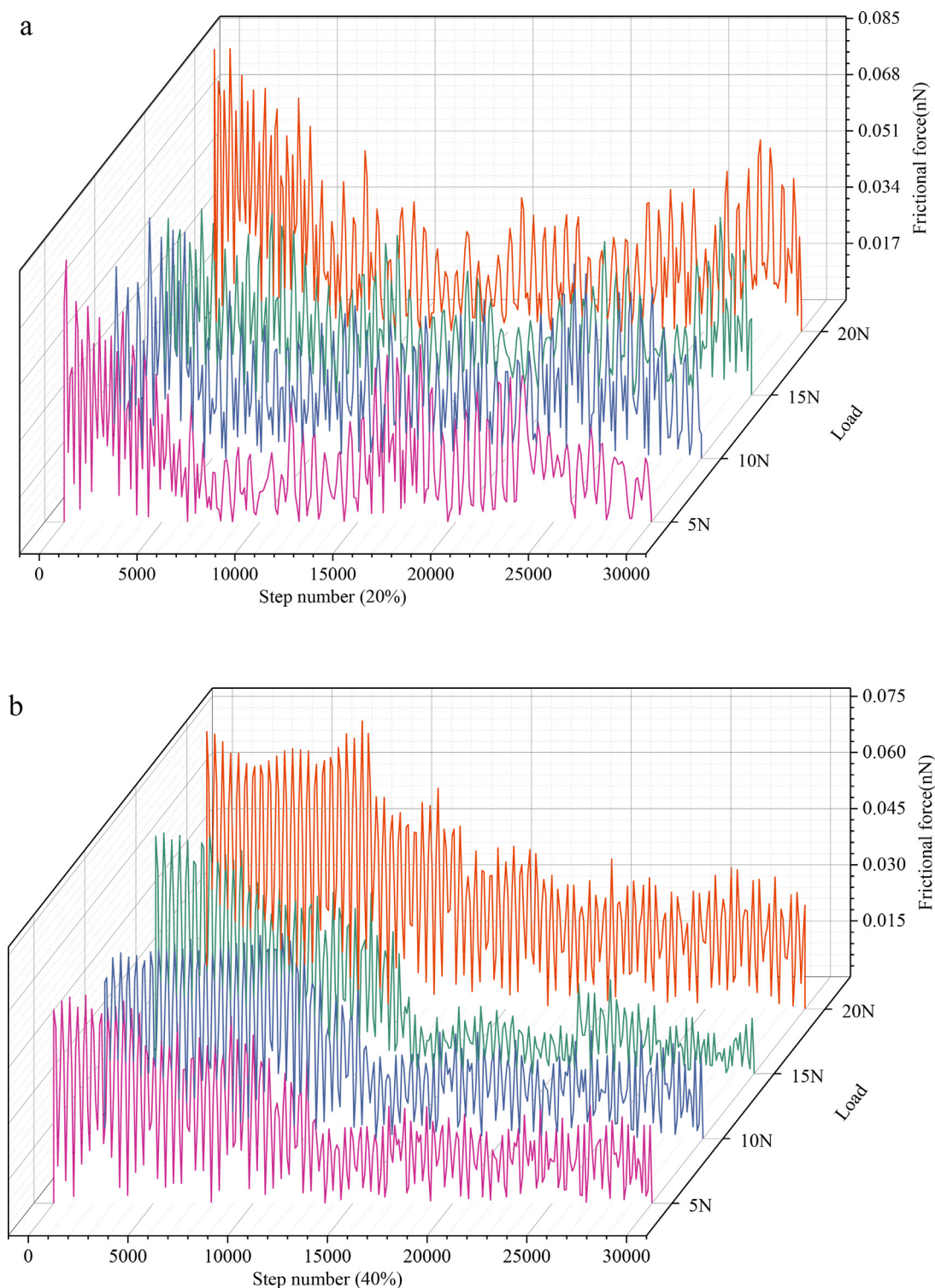


Fig. 5. The change of friction force under the influence of load of 5 N, 10 N, 15 N and 20 N with different water content. (a) 20% water content; (b) 40% water content; (c) 60% water content; (d) 80% water content.

timized after modeling, and in the experiment, the temperature is controlled rather than the energy, and there is a clear atomic number, pressure, and temperature environment in the model. Therefore, the NPT ensemble was selected in this study. Among them, the annealing calculation is carried out between 300 and 800 K, and the maximum number of iterations of geometric optimization is set to 500,000.

2.2.3. Selection of geometric optimization calculation method

Geometric optimization is a key premise of this MD simulation study, and the algorithm selection for geometric optimization is crucial. The steepest descent algorithm is a linear search along the potential energy surface's energy decrease gradient, which is suited for structural optimization with a considerable divergence from equilibrium. The conjugate gradient is conjugate in the direction

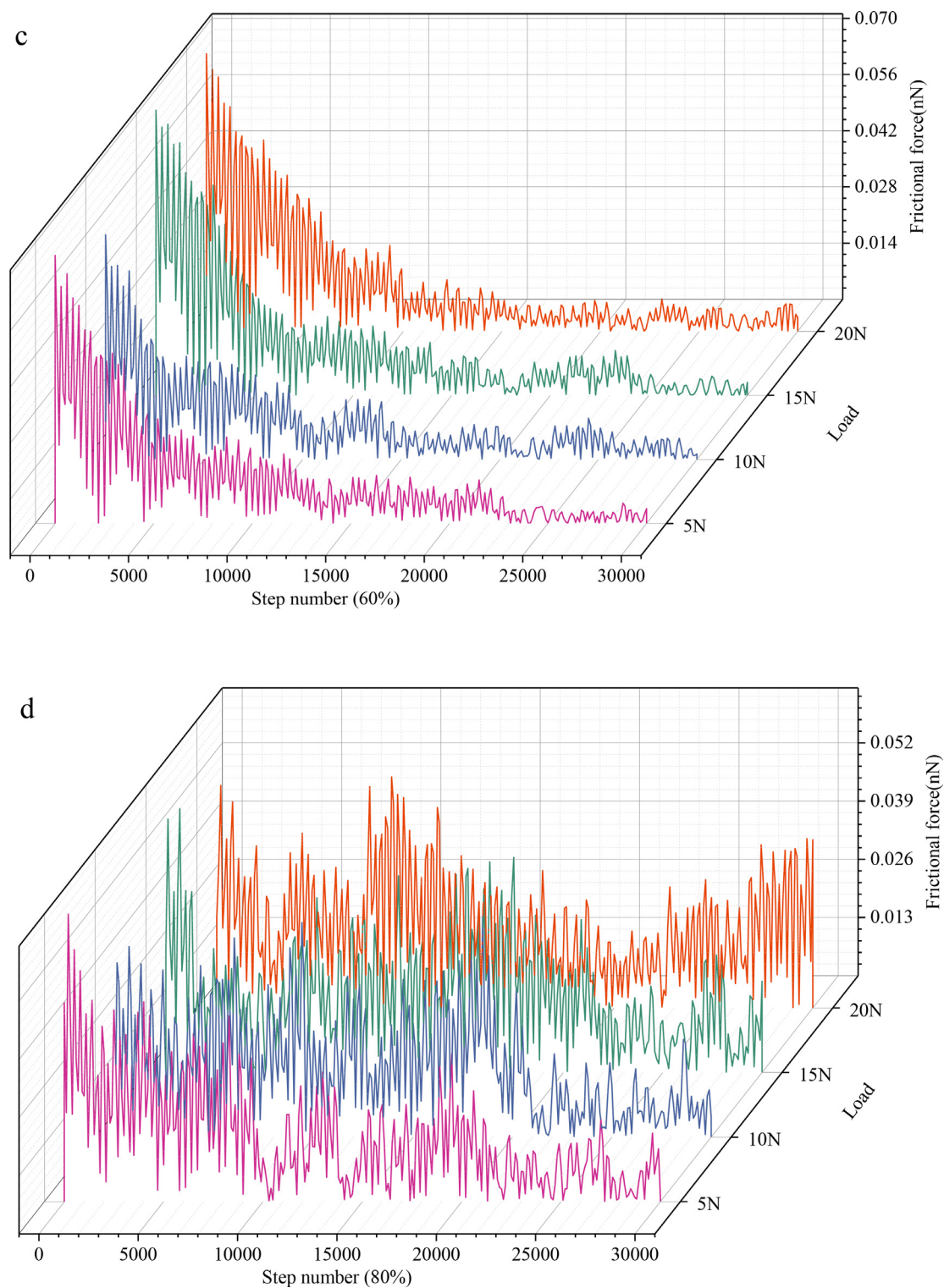


Fig. 5. Continued

of the search energy gradient across the algorithm, which is excellent for optimizing structures close to equilibrium. The Newton-Raphson algorithm, which includes Quasi-Newton and ABNR, predicts energy minima by combining gradient and curvature functions, making it suited for the optimization of structures near equilibrium. The Smart algorithm is a combination of the preceding techniques that is fine-tuned using the Conjugate gradient algorithm or the Newton-Raphson algorithm after being optimized using

the Steepest descent algorithm. Smart algorithm is the best and most comprehensive algorithm in this comparison, hence Smart algorithm is the ultimate choice.

2.2.4. Calculate

In this paper, in order to ensure that there is no accidental calculation error. All mechanical and tribological calculations were performed three times and then averaged. Compass force field is

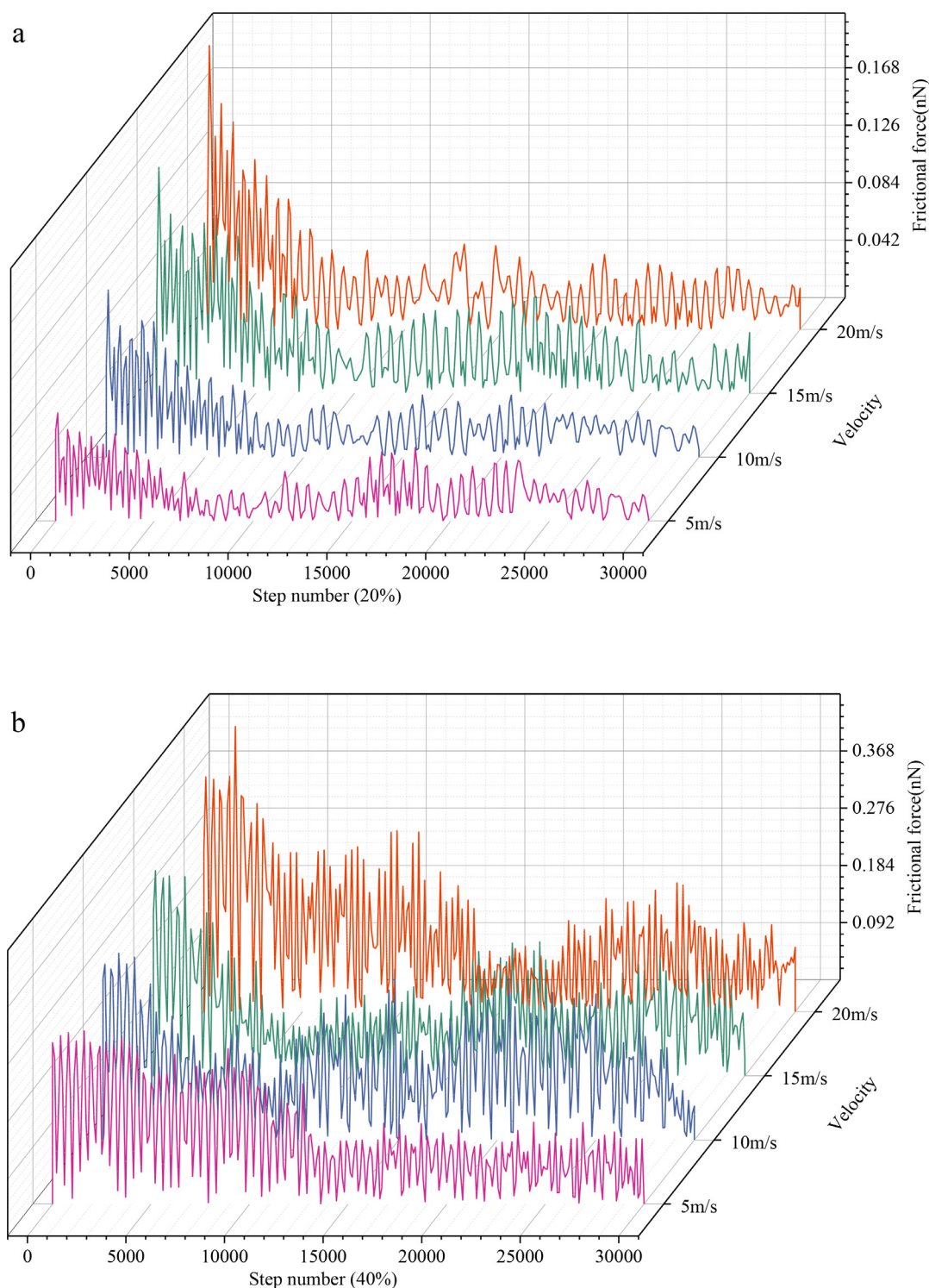


Fig. 6. The change of friction force of hydrogel model with different water content under the influence of friction speed of 5, 10, 15 and 20 m/s. (a) 20% water content; (b) 40% water content; (c) 60% water content; (d) 80% water content.

used in mechanical calculations. In the friction calculation, the model is firstly subjected to relaxation calculation to obtain a stable structure under a specific atmospheric pressure. An iron ball with A radius of 8 Å was placed in the center of the hydrogel model and the friction force was obtained by moving the ball. The speed unit is Å per step, and the step size is 0.5fs.

3. Results

3.1. Spatial distribution

In this simulation, all N(HA) atoms and O(H₂O) atoms in the amorphous model were selected in the Edit Sets module. Perform

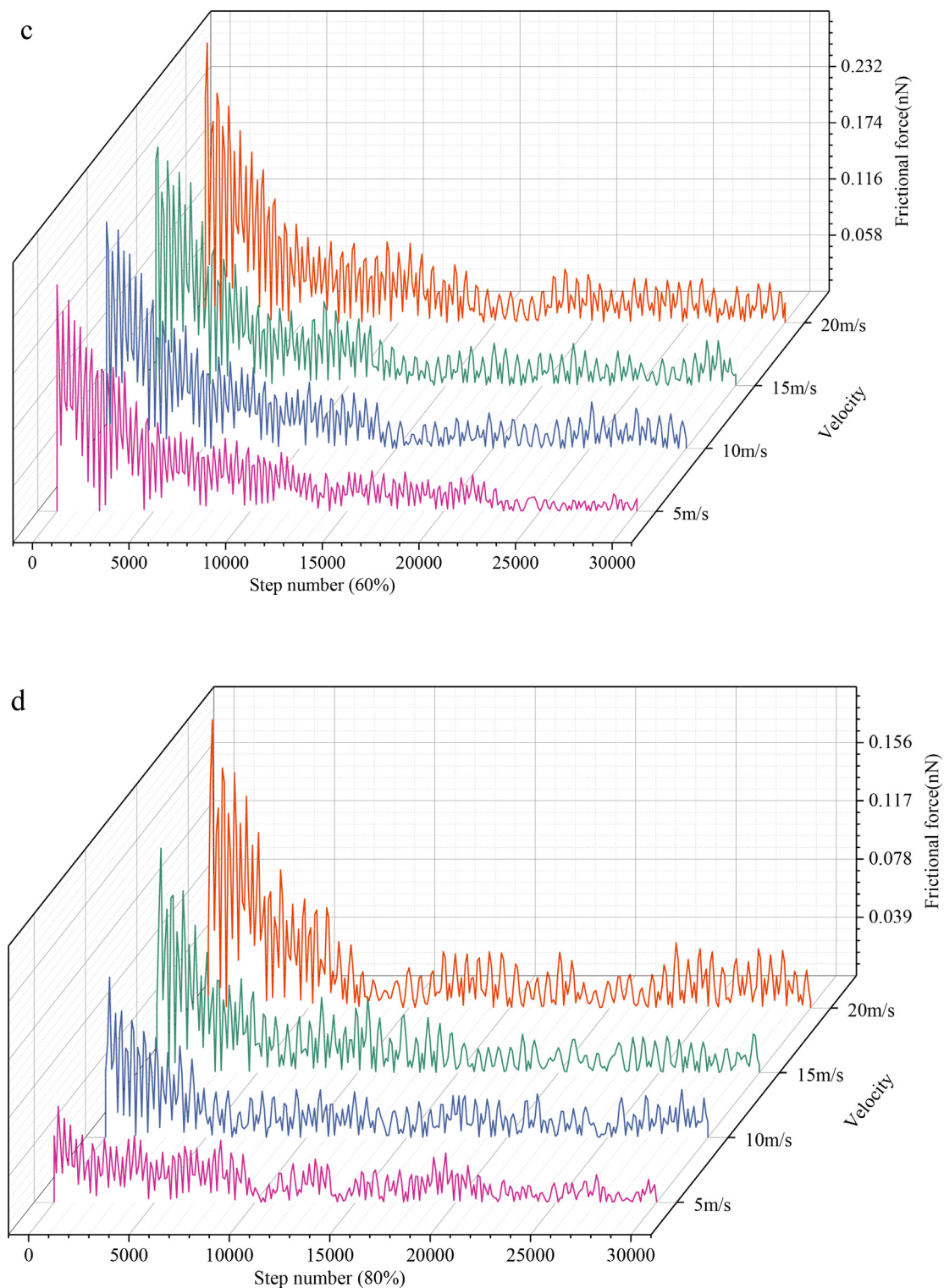


Fig. 6. Continued

radial distribution function analysis in Forcite module. In the simulation calculation of different percentage compression, the 20 and 40% water content models expand farther outward when they are originally loose, the radial distribution function (RDF) thereby decreases continuously with the compression. However, on account of the increasing intermolecular force in the models with 60 and 80% water content, the higher the uniaxial compression of the model, the more water molecules surround the N atoms in the HA.

The RDF, therefore, increases with the increase of the compression percentage, and this trend is most evident in the case of 80% water content.

3.2. Wear resistance

Fig. 4 depicts the arrangement and stacking of atoms during the friction simulation process. The friction contact point between the

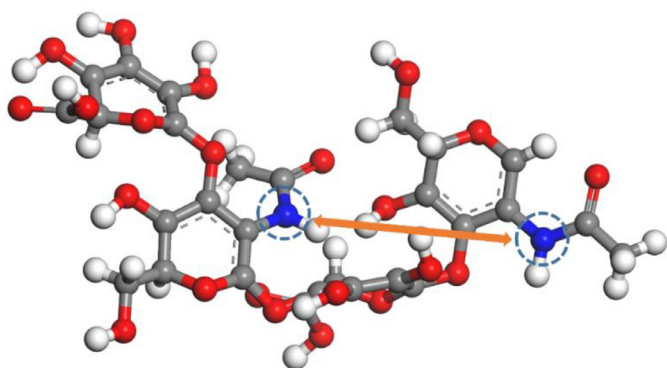


Fig. 7. The distance between the N atoms of two HA molecular units during the stress-strain process is shown.

friction ball and the hydrogel is represented by the dark blue atom block in the Z image. The atomic motion of 3, 6, 9, 12, and 15 ps during the friction process is presented. The friction ball creates preliminary friction on the hydrogel in 3 ps, and the atom stacking at the margin of the friction contact point is minimal. Certain elements of the model react subtly to the forces of friction

during friction due to intermolecular forces, causing the height of the atoms to increase and decrease. The phenomena of fluctuating atomic heights become more prevalent in the 6 and 9 ps. When the atomic height of the entire model surface reaches 12 ps, there is already a distinct color difference, which is comparable to the presence of a clear height dividing line.

3.2.1. Load

The load represents the pressure maintained by the friction ball against the hydrogel model in the friction environment. As shown in Fig. 5, the friction forces of hydrogel models with different water content under different loads are simulated and analyzed. In this study, the simulation calculation is carried out at a certain speed under the friction environment of 5 N, 10 N, 15 N, and 20 N, respectively. The step size is set to 0.5fs, and each step is 1 Å. As evident from Fig. 5, the friction force of the hydrogel model with a certain water content increases with the increasing pressure, which corroborates with the traditional mechanical friction law. In addition, with the increase of water content, the maximum friction of the model also tends to decrease. When the water content is 20%, 40%, 60%, 80%, the maximum friction is 0.068~0.085 nN, 0.060~0.075 nN, 0.056 ~ 0.070 nN, 0.039 ~ 0.052 nN respectively. It can be seen that in the range of 20–80% water content, wear resistance enhances with the increase in water content.

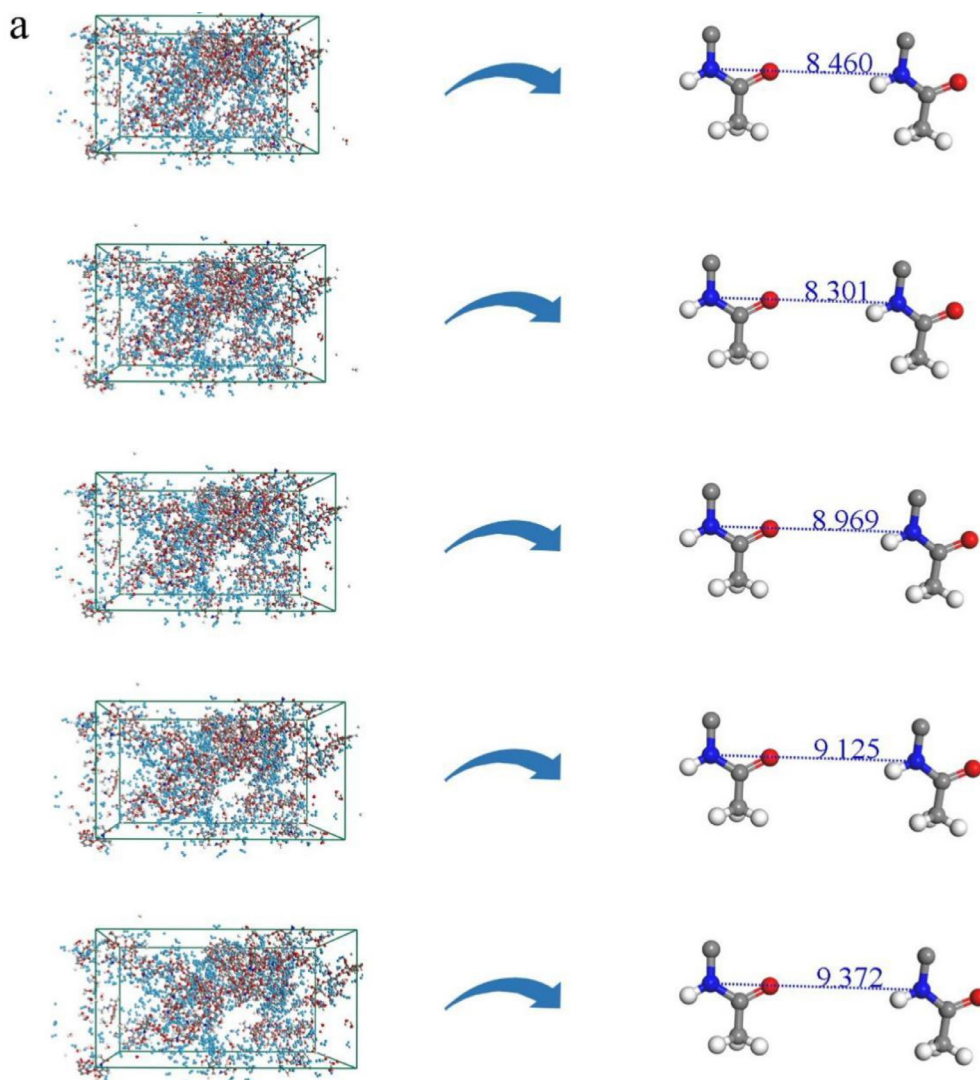


Fig. 8. Stress-strain simulation of different water content. (a) the structural state diagram of the model under uniaxial tension; (b) the stress-strain data curve.

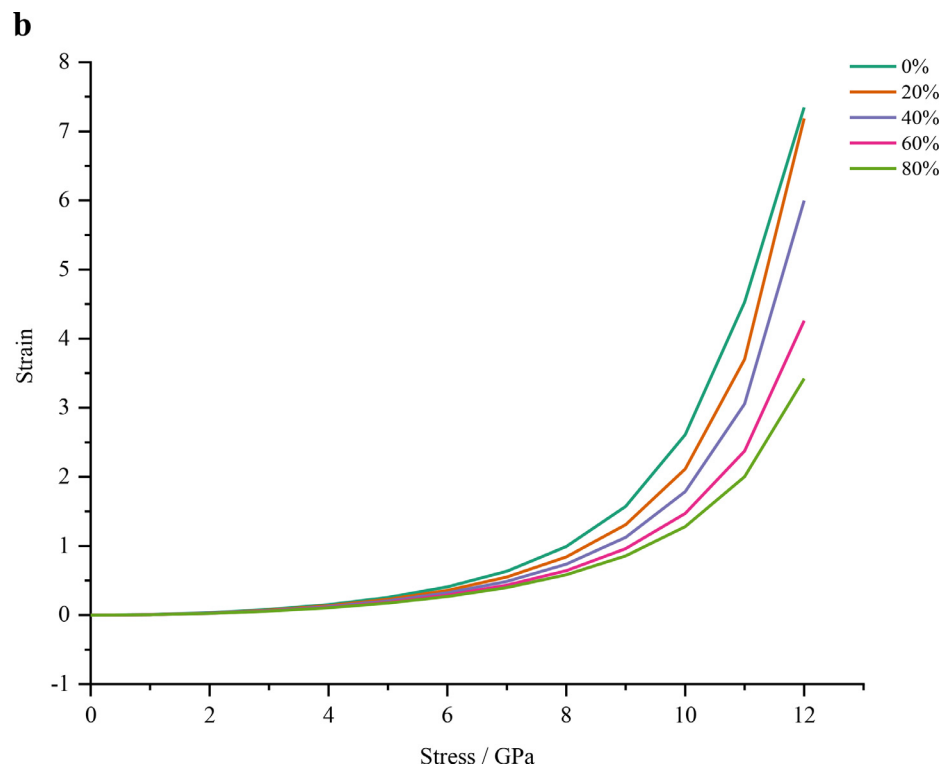


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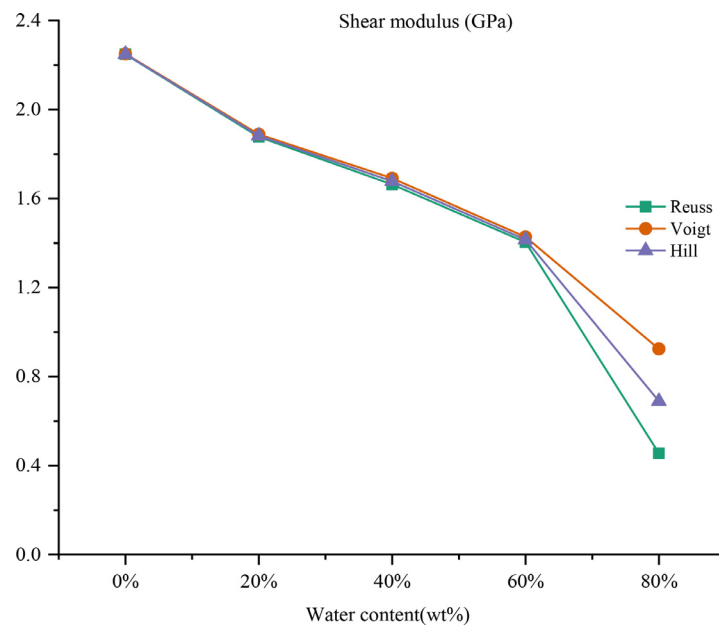


Fig. 9. Shear modulus parameters of hydrogel models with different water content.

3.2.2. Speed

The friction force of the hydrogel model with variable water content under the influence of different speeds is calculated and analyzed in Fig. 6. The load is held constant in this investigation, and the simulation calculations are performed at speeds of 5 m/s, 10 m/s, 15 m/s, and 20 m/s, respectively. Each step is 1 Å and the step size is set to 0.5fs. According to the data in Fig. 6, the friction force of the network structure with various water content increases as the friction speed increases. The principal cause is the sudden increase in the squeezing pressure on atoms and molecules as the speed increases, causing more friction to occur. From the effect

of water content, the maximum friction force is between 0.168–0.200 nN when the water content is 20%. At 40%, 60% and 80% water content, the maximum value of friction is between 0.368–0.400 nN, 0.232 and 0.300 nN and 0.156–0.200 nN respectively. The result of this data is the same as the change trend data impacted by different loads: the friction force drops gradually as the water content increases, and the wear resistance is best at 80%. The maximum friction force increases from 20 to 40% water content, and we anticipate that this is on account of the inadequate intergrading force at 20%, which leads to a loose structure of the hydrogel and extremely weak mechanical characteristics, the friction

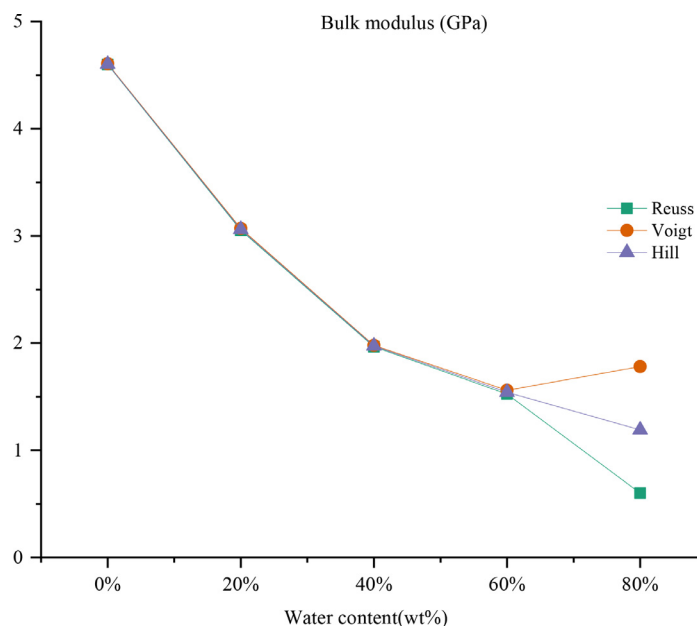


Fig. 10. Bulk modulus parameters of hydrogel models with different water content.

force created during the impact is, thus, not strong. This agrees with the early stage examination of the spatial structure.

3.3. Mechanical properties

3.3.1. Stress-strain

The uniaxial stress-strain of a HA hydrogel model with variable water content was studied using MD simulation. Fig. 8a shows how the model structure and the spacing between the N atoms of neighboring repeating units in the HA molecular chain changed during the uniaxial tensile simulation. The length of the hydrogel is continually lengthened as the one-way force is steadily applied, and the state of the molecular chain is gradually changed. As illustrated in Fig. 7, the distance between N atoms in adjacent molecular units in the hydrogel molecular chain was measured. The distance between adjacent N atoms is constantly rising, which corresponds to the physical state of stretch. However, the distance between N atoms decreases at some of these instances, contradicting the overall increasing trend. The main reason for this is that each atom moves randomly during stress-strain MD simulation, and dynamic rearrangement between molecules occurs constantly. As a result, the relative decrease in distance between the measured atoms is a normal phenomenon during stretching, and the overall change trend will not change.

Fig. 8b shows that when the uniaxial force increases, the strain of the hydrogel model with varied water content increases. The hydrogel with 0% water content is the first to manifest strain, and the strain develops rapidly as the applied force increases, with the anti-deformation ability being quite poor. The hydrogel model with 80% water content is the last to produce strain when compared to other hydrogel models with different water content, and also has the best anti-deformation ability. In general, the force required by the hydrogel model with water content ranging from 0% to 80% increases gradually at the start of strain, and the anti-deformation ability improves over time. The reason governing this behavior is the increase in the number of water molecules organizing around the HA molecular chain as the number of water molecules in the model grows. The force between the water molecule and the HA hydrogel molecular chain thereby grows, improving the anti-deformation capacity of the complete model.

3.3.2. Shear modulus

Shear modulus is one of the important mechanical properties to study the properties of hydrogels with different water content, which can demonstrate the shear strain resistance of the materials. The rigidity of the material increases with the increase of shear modulus. In addition, the reciprocal of the shear modulus is the shear flexibility, which is used to indicate the difficulty of the shear deformation of the material.

As illustrated in Fig. 9, the mechanical properties of hydrogel models with varied water content are studied and determined using MD simulation. Fig. 9 shows that when the water content is 0%, the difference between each parameter is substantial, but when the water content is 20%, the difference is extremely minor. The parameters of the shear modulus decrease as the proportion of water molecules in the hydrogel increases, and the shear modulus reaches its minimum at 80 wt%.

3.3.3. Bulk modulus

When a certain position of the material is under pressure, the bulk modulus is a stable material constant that can reflect the degree of deformation around the compression position. As shown in Fig. 10, we also employed MD simulation to analyze and calculate the bulk modulus of hydrogel models with different water content. The results show that the changing trend is similar to the shear modulus, and that the values of various parameters gradually decrease as the number of water molecules increases, indicating better deformation properties, and that the hydrogel containing 80wt% water molecules displays a better compression resistance.

3.3.4. Young modulus and Poisson ratios

Young modulus can also characterize the ability of materials to resist deformation, which is one of the important properties in elastic material mechanics and is highly significant for the study of metallic materials, semiconductors, nanomaterials, and polymers. Poisson ratios can characterize the deformation constant of hydrogel under uniaxial tension or compression. In this study, MD was used to simulate and calculate the Young modulus and Poisson ratios of the hydrogels model with different water content, as shown in Table 2. The related parameters of Young modulus and Poisson ratios rise with the increase in water molecule content, as shown

in the chart. The model's ability to resist deformation increases as the water content increases and the hydrogel becomes less prone to plastic deformation. We believe that this trend is mainly due to the change of the molecular ratio in the model, which leads to the change of Initial Pressure in the model, which leads to the deviation of these parameters from the original downward trend. It can be concluded that Initial Pressure is a very important factor affecting the simulation results.

4. Discussion

In this work, molecular dynamics simulations were used to study the effects of water content on the spatial distribution, tribological and mechanical properties of hyaluronic acid hydrogel networks. In this paper, the radial distribution functions of 20, 40, 60 and 80% water content models and the friction and mechanical parameters under different loads and friction velocities are analyzed and calculated.

The radial distribution function is a correlation function that uses the coordinates of one atom to compute the distribution probability of another atom within the equivalent radius. In the simulation computation process, it can represent the spatial distribution of individual atoms or molecules.

Based on MD simulation, the radial distribution function (RDF) of N (HA)-O (H₂O) is calculated under uniaxial compression of 10, 20, 30, and 40% of different water content models, as shown in Figs. 2 and 3. Fig. 3 shows that the RDF falls between 20 and 40% water content, and then increases dramatically between 60 and 80% water content. And the RDF of each model after 10%, 20%, 30%, and 40% compression strain demonstrates that the RDF of 20 and 40% water content models falls gradually as compression percentage increases, whereas the RDF of 60 and 80% water content models grows gradually as compression percentage increases. The fundamental reason is that, at 20 and 40% water content, the gravitational attraction between the HA molecular chain and water molecules is relatively weak, and the mechanical characteristics are not high, thus the model is loose, and RDF decreases as water molecule concentration increases. While the water content begins at 60% Mel 80%, the gravitational attraction between the HA molecular chain and the water molecules steadily strengthens, and the mechanical qualities improve. The RDF in this region rises as the water content rises, indicating that the N atoms in the HA polymer chain are gradually surrounded by more water molecules.

Kanca et al. [39] used a self-made friction tester to evaluate the tribological properties of composite hydrogels in vitro. Murakami et al. [40] used finite element analysis to study the effect of the aging process of freeze-thawing hydrogel on its friction behavior. Wu et al. [41] added graphene to the alginate saline gel to enhance the friction and wear resistance of the hydrogel. Arjmandi et al. [42] conducted an analysis of the tribological properties of interpenetrating polymer network based on alginate-polyacrylamide hydrogels for artificial cartilage. Blum et al. [43] made a detailed analysis of the mechanical and tribological properties of this new composite hydrogel and compared it with natural cartilage tissue. Joint disease has long been a disturbing condition, and because patients do not want to rely on total joint replacement surgery, researchers in the biomedical sector have been researching alternative remedies. Since the widespread usage of hydrogel, the demand for its customization for the treatment of joint problems has grown exponentially, with wear resistance being one of the most important requirements. MD simulation can be utilised to express the influence of macroscopic characteristics on molecule and atom structural changes in this context. The friction force of a hydrogel model with different water content is simulated

in this paper under the impact of different loads and friction speeds.

During friction, the atomic stacking around the friction point has appeared as a result of the long-term buildup of friction, and the atomic extrusion fluctuation generated by the intermolecular force on the friction surface has become more visible. When an atom is squeezed by friction, the intermolecular force is greatest near the contact point, therefore the atom moves swiftly outward along the vertical direction due to the sudden rise in force near the contact point. Because the intermolecular force is not confined to the friction point in the friction simulation, other places of the model will also receive a considerably lower force than the friction contact site extrusion force, resulting in sporadic atomic ups and downs. And, according to the friction simulation data curve under the influence of different loads and friction speeds, the friction force increases with the increase of the load, and increases with the increase of the friction speed. Whether it is the effect of load or the effect of friction speed, the friction force decreases with the increase of water content. More importantly, the shear strain, bulk modulus, related parameters of the hydrogels decreased with increasing water content. However, Young's modulus is on the contrary, and we think that changing the molecular ratio in the model leads to the change of Initial Pressure.

In this paper, the effects of different water content on the spatial distribution, mechanical properties and tribological mechanism of hydrogels were analyzed in detail. It fills the gap in the current simulation research.

5. Conclusions

In this work, molecular dynamics simulation was employed for the investigation of the effects of varying water content on the spatial distribution, tribological and mechanical properties of the hyaluronic acid hydrogel network. This investigation focuses on the radial distribution function of 20, 40, 60, and 80% water content model under different compression percentages, the friction force under different load and friction speed, and the mechanical parameters of different water content.

The results show that the spatial distribution state of the hydrogel networks with water contents of 20 and 40% is poor, and the network model exhibits relatively loose properties without external force. When the water content reaches 60–80%, the first peak of the radial distribution function increases with the increase of the water content, and gradually increases with the increase of the compression, indicating that the water content is 60–80%. The intermolecular forces of the gel network gradually increased, and the network model became compact.

In the calculation of friction force under the influence of different friction environments when the friction speed is constant, the friction force increases with the increase of load, and the maximum friction force decreases with the increase of water content. With a fixed load, the friction force increases with the increase of friction speed, and the maximum friction force also decreases with the increase of water content. In the determination of mechanical characteristics, the calculation parameters of each modulus decrease as the water content increases. When the water content is between 20 and 80%, the wear resistance and mechanical properties of hyaluronic acid hydrogel are generally improved with the increase of water content. The hydrogel network, which has an 80% water content, demonstrates the optimum performance.

Declaration of Competing Interest

The authors declare that they have no conflict of interest.

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Reference

- [1] J.K. Roy, H.P. Pinto, J. Leszczynski, Interaction of epoxy-based hydrogels and water: a molecular dynamics simulation study, *J. Mol. Graph. Model.* 106 (2021) 107915.
- [2] H. Gao, C. Yu, Q. Li, et al., Injectable DMEM-induced phenylboronic acid-modified hyaluronic acid self-crosslinking hydrogel for potential applications in tissue repair, *Carbohydr. Polym.* 258 (2021) 117663.
- [3] H. Ma, J. Yu, L. Liu, et al., An optimized preparation of nanofiber hydrogels derived from natural carbohydrate polymers and their drug release capacity under different pH surroundings, *Carbohydr. Polym.* 265 (2021) 118008.
- [4] N. Gull, S.M. Khan, O.M. Butt, et al., Inflammation targeted chitosan-based hydrogel for controlled release of diclofenac sodium, *Int. J. Biol. Macromol.* 162 (2020) 175–187.
- [5] S. a P. Siboro, D.S.B. Anugrah, K. Ramesh, et al., Tunable porosity of covalently crosslinked alginate-based hydrogels and its significance in drug release behavior, *Carbohydr. Polym.* 260 (2021) 117779.
- [6] J. Wang, R. Youngblood, L. Cassinotti, et al., An injectable PEG hydrogel controlling neurotrophin-3 release by affinity peptides, *J. Control. Release* 330 (2021) 575–586.
- [7] Y. Yang, Z. Zhu, R. Gao, et al., Controlled release of MSC-derived small extracellular vesicles by an injectable Diels-Alder crosslinked hyaluronic acid/PEG hydrogel for osteoarthritis improvement, *Acta Biomater.* 128 (2021) 163–174.
- [8] M. Ku, S. Cheung, W. Slattery, et al., An extended release ciprofloxacin/dexamethasone hydrogel for otitis media, *Int. J. Pediatr. Otorhinolaryngol.* 138 (2020) 110311.
- [9] C. Hwang, S.Y. Lee, H.J. Kim, et al., Polypseudorotaxane and polydopamine linkage-based hyaluronic acid hydrogel network with a single syringe injection for sustained drug delivery, *Carbohydr. Polym.* 266 (2021) 118104.
- [10] B. Teng, S. Zhang, J. Pan, et al., A chondrogenesis induction system based on a functionalized hyaluronic acid hydrogel sequentially promoting hMSC proliferation, condensation, differentiation, and matrix deposition, *Acta Biomater.* 122 (2021) 145–159.
- [11] Y. Dong, M. Cui, J. Qu, et al., Conformable hyaluronic acid hydrogel delivers adipose-derived stem cells and promotes regeneration of burn injury, *Acta Biomater.* 108 (2020) 56–66.
- [12] M.S. Bae, J.Y. Ohe, J.B. Lee, et al., Photo-cured hyaluronic acid-based hydrogels containing growth and differentiation factor 5 (GDF-5) for bone tissue regeneration, *Bone* 59 (2014) 189–198.
- [13] S. Li, M. Pei, T. Wan, et al., Self-healing hyaluronic acid hydrogels based on dynamic Schiff base linkages as biomaterials, *Carbohydr. Polym.* 250 (2020) 116922.
- [14] L. Koivusalo, M. Kauppi, S. Samanta, et al., Tissue adhesive hyaluronic acid hydrogels for sutureless stem cell delivery and regeneration of corneal epithelium and stroma, *Biomaterials* 225 (2019) 119516.
- [15] H. Kim, W.G. Koh, H.J. Lee, Effects of basic fibroblast growth factor combined with an injectable in situ crosslinked hyaluronic acid hydrogel for a dermal filler, *React. Funct. Polym.* 164 (2021).
- [16] G. Agarwal, S. Agiwal, A. Srivastava, Hyaluronic acid containing scaffolds ameliorate stem cell function for tissue repair and regeneration, *Int. J. Biol. Macromol.* 165 (Pt A) (2020) 388–401.
- [17] A. Gilarska, J. Lewandowska-Lancucka, W. Horak, et al., Collagen/chitosan/hyaluronic acid - based injectable hydrogels for tissue engineering applications - design, physicochemical and biological characterization, *Colloids Surf. B Biointerfaces* 170 (2018) 152–162.
- [18] J. Yang, Z. Tang, Y. Liu, et al., Comparison of chondro-inductivity between collagen and hyaluronic acid hydrogel based on chemical/physical microenvironment, *Int. J. Biol. Macromol.* 182 (2021) 1941–1952.
- [19] P. Zhai, X. Peng, B. Li, et al., The application of hyaluronic acid in bone regeneration, *Int. J. Biol. Macromol.* 151 (2020) 1224–1239.
- [20] S.W. Kang, J.S. Kim, K.S. Park, et al., Surface modification with fibrin/hyaluronic acid hydrogel on solid-free form-based scaffolds followed by BMP-2 loading to enhance bone regeneration, *Bone* 48 (2) (2011) 298–306.
- [21] Y.M. Tolba, S.S. Omar, D.A. Nagui, et al., Effect of high molecular weight hyaluronic acid in treatment of osteoarthritic temporomandibular joints of rats, *Arch. Oral Biol.* 110 (2020) 104618.
- [22] Z. Cai, Y. Feng, C. Li, et al., Magnoflorine with hyaluronic acid gel promotes subchondral bone regeneration and attenuates cartilage degeneration in early osteoarthritis, *Bone* 116 (2018) 266–278.
- [23] J.H. Seo, S.Y. Lee, S. Kim, et al., Monopotassium phosphate-reinforced *in situ* forming injectable hyaluronic acid hydrogels for subcutaneous injection, *Int. J. Biol. Macromol.* 163 (2020) 2134–2144.
- [24] W. Kong, Y. Gao, Q. Liu, et al., The effects of chemical crosslinking manners on the physical properties and biocompatibility of collagen type I/hyaluronic acid composite hydrogels, *Int. J. Biol. Macromol.* 160 (2020) 1201–1211.
- [25] L.S. Relleve, A.K.R. Gallardo, M.G. Tecson, et al., Biocompatible hydrogels of carboxymethyl hyaluronic acid prepared by radiation-induced crosslinking, *Radiat. Phys. Chem.* (2021) 179.
- [26] P.C. Dromel, D. Singh, T. Christoff-Tempesta, et al., Controlling growth factor diffusion by modulating water content in injectable hydrogels, *Tissue Eng. Part A* 27 (11–12) (2021) 714–723.
- [27] A. Vesković, Đ. Nakarada, A. Popović Bijelić, A novel methodology for hydrogel water content determination by EPR: the basis for real-time monitoring of controlled drug release and hydrogel swelling and degradation, *Polym. Test.* 98 (2021).
- [28] V.T. Tran, M.T.I. Mredha, I. Jeon, High-water-content hydrogels exhibiting superior stiffness, strength, and toughness, *Extreme Mech. Lett.* 37 (2020).
- [29] D. Sun, J. Zhou, Effect of water content on microstructures and oxygen permeation in PSiMA-IPN-PMPC hydrogel: a molecular simulation study, *Chem. Eng. Sci.* 78 (2012) 236–245.
- [30] M. Yuan, W. Nie, H. Yu, et al., Experimental and molecular dynamics simulation study of the effect of different surfactants on the wettability of low-rank coal, *J. Environ. Chem. Eng.* 9 (5) (2021).
- [31] O. Bertran, C. Saldías, D.D. Díaz, et al., Molecular dynamics simulations on self-healing behavior of ionene polymer-based nanostructured hydrogels, *Polymer* 211 (2020).
- [32] D. Hou, J. Xu, Y. Zhang, et al., Insights into the molecular structure and reinforcement mechanism of the hydrogel-cement nanocomposite: an experimental and molecular dynamics study, *Compos. Part B Eng.* 177 (2019).
- [33] S.G. Lee, G.F. Brunello, S.S. Jang, et al., Molecular dynamics simulation study of P (VP-co-HEMA) hydrogels: effect of water content on equilibrium structures and mechanical properties, *Biomaterials* 30 (30) (2009) 6130–6141.
- [34] M. Rezaeisadat, A.K. Bordbar, R. Omidyan, Molecular dynamics simulation study of curcumin interaction with nano-micelle of PNIPAAm-b-PEG co-polymer as a smart efficient drug delivery system, *J. Mol. Liq.* 332 (2021).
- [35] B. Wu, M.E. Wiseman, M.E. Seitz, et al., Impact of morphology on O₂ permeability in silicone hydrogel membranes: new insights into domain percolation from experiments and simulations, *J. Membr. Sci.* 621 (2021).
- [36] K. Luo, N. Yudewitz, G. Subhash, et al., Effect of water concentration on the shock response of polyethylene glycol diacrylate (PEGDA) hydrogels: a molecular dynamics study, *J. Mech. Behav. Biomed. Mater.* 90 (2019) 30–39.
- [37] J. Zidek, A. Milchev, J. Jancar, et al., Deformation-induced damage and recovery in model hydrogels – A molecular dynamics simulation, *J. Mech. Phys. Solids* 94 (2016) 372–387.
- [38] P. Taweechat, R.B. Pandey, P. Sompornpisut, Conformation, flexibility and hydration of hyaluronic acid by molecular dynamics simulations, *Carbohydr. Res.* 493 (2020) 108026.
- [39] Y. Kanca, P. Milner, D. Dini, et al., Tribological properties of PVA/PVP blend hydrogels against articular cartilage, *J. Mech. Behav. Biomed. Mater.* 78 (2018) 36–45.
- [40] T. Murakami, N. Sakai, S. Yarimitsu, et al., Evaluation of influence of changes in permeability with aging on friction and biphasic behaviors of artificial hydrogel cartilage, *Biotribology* 26 (2021).
- [41] L. Wu, Z. Zhang, M. Yang, et al., Graphene enhanced and in situ-formed alginate hydrogels for reducing friction and wear of polymers, *Colloids Surf. A Physicochem. Eng. Asp.* 589 (2020).
- [42] M. Arjmandi, M. Ramezani, A. Nand, et al., Experimental study on friction and wear properties of interpenetrating polymer network alginate-polyacrylamide hydrogels for use in minimally-invasive joint implants, *Wear* 406–407 (2018) 194–204.
- [43] M.M. Blum, T.C. Ovaert, Low friction hydrogel for articular cartilage repair: evaluation of mechanical and tribological properties in comparison with natural cartilage tissue, *Mater. Sci. Eng. C Mater. Biol. Appl.* 33 (7) (2013) 4377–4383.
- [44] H. Ju, R. Zhou, J. Luan, et al., Multilayer Mo₂N-Ag/SiN_x films for demanding applications: morphology, structure and temperature-cycling tribological properties, *Mater. Des.* 223 (2022) 111128.